

Sai Sri Rama Khandavilli

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PROFESSIONAL SUMMARY

Full Stack Developer with 5+ years of experience building scalable web applications and microservices using **Python, FastAPI, Django, React, Agentic AI, PostgreSQL** and **LLMs**. Experienced in developing **REST** and **GraphQL** APIs, securing systems with OAuth2 and JWT, and building AI-powered solutions that reduce support load and improve user experience. Strong in **TDD**, database optimisation, **CI/CD** pipelines, and Agile delivery.

PROFESSIONAL EXPERIENCE

Full Stack AI Developer | Whipcream

April 2026 – Present

- Designed and developed intelligent AI-driven applications from scratch using **Deepseek LLM**, including end-to-end system architecture and implementation.
- Built scalable, responsive user interfaces using **React**, leveraging hooks and component-driven architecture to improve performance, reusability, and user experience.
- Engineered robust backend systems using **Python FastAPI**, implementing efficient data processing workflows and integrating third-party APIs to support scalable application functionality.
- Orchestrated **agentic AI workflows** and **LLM** integrations to automate multi-step reasoning, contextual analysis, and intelligent decision-making across AI-powered applications.
- Developed AI agent workflows using **LangGraph** for multi-step orchestration and state management, leveraging **Pydantic** for structured data validation, schema enforcement, and reliable input/output handling.
- Worked directly with **2 co-founders** from ideation to MVP, **independently** converting ambiguous ideas into PRDs/specs, workflows, and 0→1 prototypes without a predefined backlog or step-by-step guidance.
- Owned architecture and key decisions across multiple integrations: introduced Nango to standardize connectors, migrated **MongoDB** → **PostgreSQL**, and recommended **TanStack** to support future streaming/real-time UI requirements.
- Led requirements/roadmaps** in a CTO-like capacity, coordinating junior developer tasks and onboarding the new CTO on scope, delivery status, and next milestones.
- Architected an **Orchestrator Agent** for diagnostic→action workflows, routing structured directives across specialized agent nodes with **conditional failure edges, structured fallbacks, confidence capping, and end-to-end audit logging** of LLM calls.
- Implemented **spec-driven development** with **Claude Code** by freezing technical contracts upfront (schemas, deterministic formulas, canonical test cases) and generating implementation + test suites from the spec, accelerating delivery while keeping builds **rapid and reproducible**.
- Designed a layered LangGraph separating **probabilistic LLM agents** (schema-constrained with **Pydantic structured outputs**) from **100% deterministic scoring nodes**, ensuring outputs remain **auditable, reproducible, and reliable** without losing AI-driven flexibility.

Full Stack Developer | Tata Consultancy Services (TCS)

Oct 2020 – Nov 2025

- Independently initiated and built an internal **GenAI chatbot** from scratch (**Python/Django**), serving 50,000+ users and offloading 1,200+ monthly queries (~35% manual effort reduction).
- Improved answer quality using **Falcon LLM + prompt refinement** over structured org knowledge (Excel/internal docs), increasing domain accuracy across high-volume support use cases.
- Managed **client/stakeholder communication** end-to-end (demos, feedback loops, deployment support) and shipped rapid fixes/releases across multi-stage production environments with minimal oversight.
- Standardised 4 portals (Chatbot / Admin / Login / API) across 30+ components by designing modular **Angular** patterns using TypeScript, **HTML5**, and **CSS3** to ensure UI consistency and reusability.
- Engineered and optimised 15+ **REST and GraphQL APIs** using **Django REST Framework**, supporting 50,000+ active users across multi-stage production environments with robust error handling and versioning.
- Optimised **PostgreSQL** schemas for 500,000+ records, achieving 40% faster retrieval by refactoring **Django ORM** query strategies and execution plans.
- Accelerated deployment frequency by 40% by architecting **Jenkins and GitLab CI/CD pipelines**, eliminating manual delivery bottlenecks across development and production environments.
- Lowered monthly cloud spend by 30% using **AWS Lambda, S3, and EC2** for rightsized, event-driven data processing workflows across all production environments.
- Contributed to a high-traffic **Java Spring Boot** microservices architecture (Java 11/17/21) supporting 50,000+ active users, applying SOLID, DDD, and Clean Architecture principles.

- Built scalable backend services using Node.js and RESTful APIs, optimising asynchronous operations to handle high-throughput requests.
- Reduced query latency by 35% via a Node.js chatbot using Microsoft Bot Framework, LUIS, QnA Maker, and Azure AI services.
- Centralised 4 data streams into 6 Power BI dashboards with custom DAX, reclaiming 75% of reporting bandwidth by replacing manual ETL processing.

AI/ML Intern | Central Tool Room & Training Centre

Jun 2019

- Benchmarked a self-driving car behavioural-cloning pipeline at 95% accuracy using CNN, TensorFlow, Keras, and OpenCV training 20,000+ samples for 25% higher stability at 30+ FPS

SKILLS

Backend : Java, Python, Node js, Spring boot, FastAPI, PostgreSQL, AWS**Frontend:** React, Tanstack, Angular**Other:** CI/CD, Jenkins, Docker, Claude Code, Vercel deployment, Agentic AI Development, Generative AI**PROJECTS**

Recipe Generator | React, FastAPI, MongoDB, OpenAI LLM, JWT Authentication, Emergent | [Link](#)

- Built a full-stack AI-powered recipe generation platform using **React, FastAPI, MongoDB, and OpenAI**, generating personalized recipes from ingredients, cuisine preferences, and dietary restrictions, supporting **12+ dietary tags** and reducing meal-planning time by **80%**.
- Developed **15+ REST API endpoints** for JWT authentication, recipe CRUD operations, search/filtering, categories, and favorites, enabling secure management of **100+ user-generated recipes**.
- Engineered structured prompt workflows and AI response validation for recipe generation, improving recommendation relevance by **30%**, reducing inconsistent outputs by **25%**, and delivering recipe responses in under **3 seconds**.

KnowledgeHub AI | TypeScript, React, TanStack Start, Supabase, pgvector, RAG, Gemini, Lovable | [Link](#)

- Built a production-ready **RAG-powered research notebook** enabling users to upload and query PDFs through semantic search, implementing document ingestion, vector embeddings, and citation-backed AI responses across **1,000+ document chunks per knowledge base**.
- Developed an end-to-end retrieval pipeline using **Supabase pgvector, text-embedding-3-small**, and hybrid context management, optimizing retrieval to the **top 6 most relevant chunks** and reducing hallucinations through grounded, citation-aware prompting.
- Engineered secure multi-user document management with **Supabase Auth**, Row-Level Security (RLS), and vector search indexing, supporting document upload, semantic retrieval, conversational follow-up questions, and persistent research notebooks with source traceability.

CERTIFICATIONS & ACHIEVEMENTS

- Google Cloud Certified: Professional Machine Learning Engineer
- Deep Learning : One Fourth Labs
- Hands-on AI for real-world applications (for TCS) : IIT Kharagpur AI4ICPS I Hub Foundation
- Forage : Software Engineer Job Simulation from Hewlett Packard

EDUCATION

MBA in Business Intelligence & Artificial Intelligence | GPA: 8.68 | Jain University | 2021–2023**B.E (Hons) in Computer Science & Engineering | GPA: 8.70 | Trident Academy of Technology | 2016–2020**